CS 374: Section A - Homework 4, Problem 1

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Throughout, this homework let $M = (\Sigma, Q, s, A, \delta)$ be an arbitrary DFA accepting L, with extended transition function δ^*, where $\Sigma = \{0, 1\}$ and $\Sigma' = \{0, 1, \#\}$.

Part (a)

$$\text{BackToFront}(L) = \{y^R \# x \mid x, y \in \Sigma^* \text{ and } xy \in L\}$$

is regular. Define an NFA $M' = (\Sigma', Q', s', A', \delta')$ with ε-transitions as follows.

$$\begin{aligned}
Q' &= (Q \times Q \times \{1, 2\}) \cup \{s'\} \\
s' &\text{ is an explicit state in } Q' \\
A' &= \{(h, h, 2) \mid h \in Q\} \\
\delta'(s', \varepsilon) &= \{(h, f, 1) \mid h \in Q \text{ and } f \in A\} \\
\delta'(s', a) &= \varnothing && \text{for every } a \in \Sigma' \\
\delta'((h, r, 1), \varepsilon) &= \delta'((h, q, 2), \varepsilon) = \varnothing \\
\delta'((h, r, 1), a) &= \{(h, p, 1) \mid p \in Q \text{ and } \delta(p, a) = r\} && \text{for every } a \in \Sigma \\
\delta'((h, r, 1), \#) &= \begin{cases} \{(h, s, 2)\} & \text{if } r = h \\ \varnothing & \text{otherwise} \end{cases} \\
\delta'((h, q, 2), a) &= \{(h, \delta(q, a), 2)\} && \text{for every } a \in \Sigma \\
\delta'((h, q, 2), \#) &= \varnothing
\end{aligned}$$

Why this is correct. M' runs two copies of M on the two halves of xy, in the order the input presents them: the suffix y arrives first and reversed, so M' runs M backwards on it, and the prefix x arrives second and in order, so M' runs M forwards on it.

- The start state s' guesses, without reading any input, the state h that M reaches after the prefix x and an accepting state $f \in A$ that M reaches at the end of xy.
- A state $(h, r, 1)$ means: the guess is h, and $\delta^*(r, u^R) = f$ for the portion u of the input read so far. Reading a moves from r to any p with $\delta(p, a) = r$, one step of M in reverse.
- The transition on # fires only when $r = h$; the input read so far is then exactly y^R, so this test says $\delta^*(h, y) = f \in A$. Only one # is ever read, since states with third component 2 have no # transition.

- A state $(h, q, 2)$ means $\delta^*(s, v) = q$ for the portion v read since the $\#$, and M' accepts only at $(h, h, 2)$.

So M' accepts w exactly when $w = y^R \# x$ for strings $x, y \in \Sigma^*$ and states $h \in Q$, $f \in A$ with $\delta^*(s, x) = h$ and $\delta^*(h, y) = f$. Since $\delta^*(s, xy) = \delta^*(h, y) = f \in A$, that happens exactly when $xy \in L$, so $L(M') = \text{BackToFront}(L)$ and this language is regular.

Sources and collaborators: None.

LLM use: I used Claude (Anthropic) to draft these constructions and explain their structure to me. I checked every construction by hand against the examples in the handout and against several strings inside and outside each language, and typeset the submission myself.

Part (b)

$$\text{FrontToBack}(L) = \{yx^R \mid x, y \in \Sigma^* \text{ and } xy \in L\}$$

is regular. There is no separator this time, so the machine must guess where y ends and x^R begins, and the two halves arrive in the opposite order from part (a): the suffix y comes first and in order, and the prefix x comes last and reversed. Define an NFA $M'' = (\Sigma', Q'', s'', A'', \delta'')$ with ε-transitions as follows.

$$\begin{aligned}
Q'' &= (Q \times Q \times \{1,2\}) \cup \{s''\} \\
s'' &\text{ is an explicit state in } Q'' \\
A'' &= \{(h, s, 2) \mid h \in Q\} \\
\delta''(s'', \varepsilon) &= \{(h, h, 1) \mid h \in Q\} \\
\delta''(s'', a) &= \varnothing && \text{for every } a \in \Sigma' \\
\delta''((h, q, 1), \varepsilon) &= \begin{cases} \{(h, h, 2)\} & \text{if } q \in A \\ \varnothing & \text{otherwise} \end{cases} \\
\delta''((h, q, 1), a) &= \{(h, \delta(q, a), 1)\} && \text{for every } a \in \Sigma \\
\delta''((h, r, 2), \varepsilon) &= \varnothing \\
\delta''((h, r, 2), a) &= \{(h, p, 2) \mid p \in Q \text{ and } \delta(p, a) = r\} && \text{for every } a \in \Sigma \\
\delta''(z, \#) &= \varnothing && \text{for every } z \in Q''
\end{aligned}$$

Why this is correct. Again M'' runs two copies of M, one on each half of the word $xy \in L$, in the order the input presents them, but now the forwards copy runs first and the backwards copy runs second.

- The start state s'' guesses the state h that M reaches after the prefix x and starts the first copy of M there. The guess h sits in the first component of every state and is never changed.
- A state $(h, q, 1)$ means: the guess is h, and $\delta^*(h, u) = q$ for the portion u of the input read so far. This phase simulates M forwards from the guessed state h instead of from s.
- The ε-transition from $(h, q, 1)$ to $(h, h, 2)$ is the guess that the first half of the input is over. It exists only when $q \in A$, so taking it records that the portion y read so far satisfies $\delta^*(h, y) \in A$.
- A state $(h, r, 2)$ means: the guess is still h, and $\delta^*(r, v^R) = h$ for the portion v read since the ε-transition; as in part (a) this runs M in reverse. The only accepting states are the states $(h, s, 2)$, so acceptance demands $\delta^*(s, v^R) = h$.

So M'' accepts w if and only if $w = yv$ for strings $y, v \in \Sigma^*$ and a state $h \in Q$ with $\delta^*(h, y) \in A$ and $\delta^*(s, v^R) = h$. Writing $x = v^R$, this says $w = yx^R$ where $\delta^*(s, x) = h$ and $\delta^*(s, xy) = \delta^*(h, y) \in A$, that is, $xy \in L$. Hence $L(M'') = \text{FrontToBack}(L)$, which is therefore regular. No string containing $\#$ is accepted, as it should be, because $\text{FrontToBack}(L) \subseteq \Sigma^*$.

Sources and collaborators: None.

LLM use: I used Claude (Anthropic) to draft these constructions and explain their structure to me. I checked every construction by hand against the examples in the handout and against several strings inside and outside each language, and typeset the submission myself.